

The mass gap: current position of this programme

This is the synthesis of the notes written since the goal was set to a proof of the Yang–Mills existence and mass-gap conjecture. It states what has been proved here, what has been imported, what has been ruled out, and what remains, with constants explicit and each claim labelled.

The short version: the conjecture decomposes into six named statements, of which two are proved here and one is imported. The clause $m < \infty$ is closed as far as variational methods reach, and reduced to the existence of the theory plus the nontriviality of one flowed correlator. The clause $m > 0$ is untouched, and **seven routes to it have been closed by explicit computation**, in two families: those built from operator norms and spectra, and those built from the ground state of the Hamiltonian. Both families fail for reasons now stated exactly, and the second failure identifies what the Euclidean formulation supplies that the Hamiltonian one does not.

1. The decomposition

The obligations map turns “prove the mass gap” into six statements about the Kogut–Susskind Hamiltonian

$$H = \frac{\hbar c}{a} \left[\frac{g^2}{2} \sum_{\ell} (-\Delta_{\ell}) + \frac{2}{g^2} \sum_p (N - \text{Re tr } U_p) \right], \quad \Delta_{a,L} = \frac{\hbar c}{a} \delta(g; N_s, G).$$

	statement	status
T1	finite lattice: unique physical ground state, $\delta > 0$	proved here
T2	$\delta \geq \gamma(g^2/2)C_2$ for $g \geq g_0$, uniform in volume	proved here from an imported theorem
T2'	$\liminf_{N_s} \delta > 0$ for every $g > 0$	open; false for $U(1)$
T3	$\delta_{\infty}(g)/(a\Lambda_{\text{lat}}(g)) \rightarrow m/(\hbar c\Lambda) \in (0, \infty)$	open
T4	continuum infinite-volume theory with the axioms	finite-volume ultraviolet stability known
S	small volume: $\Delta = \delta_1 g^{2/3} \hbar c / L [1 + O(g^{2/3})]$	upper side proved, lower side fixed-lattice only

The conjecture is T2' together with T3, given T4, and the critical-coupling note shows that T2' is exactly the absence of a zero-temperature bulk phase transition, hence on its second-order branch the uniqueness of the continuum limit.

2. What is proved here

T1 (obligations map §2): compact resolvent, a simple strictly positive ground state, a positive gap inherited by the physical sector, for every a , N_s , g and compact connected G .

T2 in Hamiltonian form (T2 note): the electric term is classical in the Peter–Weyl partition and the magnetic term is a bounded range- $\{0, 1\}^3$ perturbation with $\beta = 48N^2/[g^4(N^2 - 1)]$, so Yarotsky’s theorem gives $\Delta_{a,L} \geq \gamma(g^2/2)C_2\hbar c/a$ for $g \geq g_0$, uniformly in the lattice size, with exponential clustering.

A Lieb–Robinson bound (note): velocity $v \leq 32eNc/g^2$, uniform in volume, because the unbounded part of H is a sum of commuting single-link Laplacians; hence a strong-coupling correlation length $\xi \leq C'aN/(\gamma C_2 g^4)$.

Upper bounds (note, Polyakov note): the Feynman–Bijl inequality with explicit constants; the strong-coupling gap pinned to order $g^2\hbar c/a$ on both sides; the abelian gap bounded by the plaquette-sine structure factor; the proof that **no operator linear in the electric field with c-number coefficients is gauge invariant for a non-abelian group**; and a necessary condition on the Polyakov-loop susceptibility.

The finiteness clause reduced (moment hierarchy): $m \leq M_{k+1}/M_k$ for every k , with the ratios decreasing, so the f -sum rule is the weakest member and the next one, $m \leq \hbar C_0(0)/\int_0^\infty C_0$, needs only T4 and $C_0 \neq 0$. The free-field value of the weakest member is $4.26 \hbar c/\sqrt{8t}$, computed in closed form and confirmed by an independent sum rule (free-field note).

Exact structural identities (note): $\Delta V = 4C_2(N|\mathcal{P}| - V)$, so the magnetic energy shifted by its Haar mean is a Laplacian eigenfunction; $|\nabla V|^2 \leq 16V$ for $SU(2)$; and an exact ground-state sum rule tying $\text{Var}_\Omega(V)$ to kinetic quantities.

Flow facts (conjugation, lattice truncation): the Wilson flow is a diffeomorphism of $G^\mathcal{E}$, unitarily implemented, so conjugation preserves the spectrum; the lattice flow carries no coupling and a doubling is dimensionless flow time $1/2$, so truncating the conjugated Hamiltonian at range Ka costs Ce^{-cK} with pure constants.

3. Seven closed routes

route	why it closes
variational upper bounds, for $m > 0$	a spectral measure with all negative moments finite can have $m = 0$
expansion around the free theory	the bound reaches m only at the confinement scale, where its coefficient is nonperturbative
blocking by projection	the boundary grows like M^2 while the block gap does not; the criterion worsens by $3M^2/8$
conjugation by the flow	the spectrum is invariant, so the gain is the truncation, which is cheap
flow before decimation	sup norms are bijection-invariant and spectra conjugation-invariant
Agmon estimate on the ground state	$\ \nabla V\ _\infty$ is extensive, so it bounds the global excess only
pointwise Gibbs domination	the comparison function bites only above the extensive threshold $V_* \simeq N \mathcal{P} $

The first five exhaust the methods built from operator norms and spectra; the last two exhaust the standard ways of extracting a local statement from the ground-state eigenvalue equation. Each closure is a computation with explicit constants.

The reason the last two fail, stated once. Every inequality derived from $-A'\Delta\Omega + (B'V - e_0)\Omega = 0$ compares the total potential with the total energy, and e_0 is extensive, so it separates configurations only by their global excess. The Euclidean weight $e^{-S_w} = \prod_p e^{-(2/g_E^2)V_p}$ factorizes over plaquettes and separates them locally by construction. That is the structural advantage of the Euclidean formulation, derived here rather than assumed.

4. Where the non-abelian structure has been isolated

Three statements distinguish the groups, and any proof of T2' must use one of them.

1. *The commutator potential* of the zero-momentum sector is identically zero for an abelian group; where present, the transverse zero-point energy confines the valley and produces the gap $\delta_1 \hbar^{4/3} g^{2/3} m^{-2/3}$ (G07, claim C133).

2. *Gauge invariance blocks the photon channel*: no c-number-coefficient operator linear in E commutes with Gauss's law for a non-abelian group, so the excitation through which the abelian gap closes must carry a Wilson line (upper-bound note Proposition 6).
3. *The one-loop valley potential* $U(a) = \frac{2\hbar c}{\pi^2 L} \Phi(La)$ vanishes identically in the abelian theory, so the holonomy is free (valley note §3b).

5. What remains

obligation	kind
T2': closedness of the gapped set	the conjecture
openness of the gapped set	technical: gap stability without frustration-freeness
T3	the conjecture, given T2'
T4	construction
local large-field control	available via the transfer-matrix
	identification: $ \Omega ^2$ is the Euclidean time-slice marginal
the large-field region itself	the growth $e^{2t\ G\ _\infty}$ is sharp , saturated by the Nielsen–Olesen mode, so the region is excluded; its entropy is beaten by its action cost by $2\pi^2/g^2$, and what remains is the constrained minimizer

The pattern across every note is uniform: **statements uniform in the cutoff are renormalization statements, and statements local in space need a local weight**. The strong-coupling theorem is uniform in the volume at fixed cutoff; the small-volume theorem is fixed-lattice; the flowed bounds are finite at fixed lattice and need the flow's renormalization in the continuum; the ground-state estimates are global because the eigenvalue equation is.

6. Assessment

The route that survives is the constructive one: conjugate by the flow, which is exact; truncate, which costs Ce^{-cK} per doubling; decimate, which is the problem; and iterate a number of times logarithmic in the scale ratio, about thirty for $SU(2)$ at a bare coupling $g_{UV}^2 = 1/2$ (blocking note §5). The decimation step requires a local large-field estimate, and the notes show that the Hamiltonian formulation cannot supply one from its own equation. The natural continuation leaves the Hamiltonian framework: identify Ω^2 with a Euclidean measure on a time slice through the transfer matrix, and work there.

What this programme has added is a map with constants: six named statements, two proved, the finiteness clause reduced to a single correlator, seven routes closed with explicit numbers, three places where the non-abelian structure is isolated in a usable form, and the exact identities and sum rule of Section 2. Several of the closures are corrections of claims made earlier in the same series, and each is recorded with the computation that forced it.